

Solving the 'Registration-to-Installation' Conversion Gap in India's Residential Solar Market

A Strategic Thesis on Decarbonizing the Last Mile

The Indian energy paradigm is currently navigating a pivotal transition, shifting from a centralized, fossil-fuel-dependent grid toward a decentralized, renewable-heavy architecture. At the center of this shift is the residential rooftop solar (RTS) segment, which represents both the most significant opportunity for democratization of energy and the most complex operational challenge in the renewable sector. The launch of the Pradhan Mantri Surya Ghar: Muft Bijli Yojana (PMSGY) in February 2024, with a monumental financial outlay of ₹75,021 crore, signaled a decisive move to install solar panels on 10 million households by the end of fiscal year 2026-2027.¹ However, as the market matures into late 2025, a stark discrepancy has emerged between the volume of registrations and the actualization of installations. This conversion gap is not merely a logistical hurdle but a multifaceted systemic crisis involving financial friction, supply chain imbalances, and regulatory inertia.

Sensemaking: Relevance, Scale, and the Urgency of the Solar Transition

The scale of India's rooftop solar ambition is unprecedented globally. The objective is to add 30 GW of residential capacity through the PMSGY scheme alone.¹ By July 2025, the country had achieved 4.9 GW of residential capacity, accounting for nearly 45% of total residential rooftop capacity.¹ While these numbers suggest progress, the underlying funnel analysis reveals a concerning trend. Despite approximately 5.79 million applications being filed on the national portal, the conversion to completed installations stands at just 22.7%.¹ This indicates that for every five households expressing high intent, only one successfully navigates the process to a functional solar system.

The urgency of addressing this gap is driven by a looming energy demand crisis. India's total installed power capacity reached 510 GW by late 2025, with solar contributing 133 GW of the mix.⁵ However, electricity demand is forecasted to double by 2030 and potentially quadruple by 2050.⁶ A significant portion of this growth is attributed to the "cooling transition," where energy demand for household air conditioners is expected to increase ninefold between 2022 and 2025.⁷ Without a robust residential solar infrastructure, this demand will continue to be met by coal-fired thermal power, which currently provides nearly 70% of generation.⁵

The Funnel Attrition and Macroeconomic Stakes

The national target of 1 crore installations by FY2027 is a critical milestone for India's 2070 Net Zero

commitment. As of December 2025, about 28 lakh households had been powered by rooftop solar, representing approximately 28% of the target, yet actual commissioned capacity under PMSGY specifically was reported at 13.1% of the target earlier in the year.³ This suggests that while adoption is accelerating, the pace remains insufficient to meet the 2027 deadline.

The economic implications for the Indian middle class are profound. The current technical potential for residential rooftop solar is estimated at over 600 GW, though economic viability is restricted to roughly 11 GW at the household level due to current tariff structures and system costs.¹⁰ Successfully closing the conversion gap would unlock significant household savings. A cost-benefit analysis of the first 6.8 lakh installations projected net savings of ₹36,089 crore over a 25-year lifecycle.¹¹

Metric of Progress	Value (as of July 2025)	Target (FY 2027)	Current Achievement (%)
Cumulative Installations (PMSGY)	1.6 Million Households	10 Million Households	16% ¹
Installed Capacity (PMSGY)	4.9 GW	30 GW	16.3% ¹
Total Applications Received	5.8 Million	N/A	N/A ¹
National Conversion Ratio	22.7%	N/A	N/A ¹
Subsidy Released (INR)	₹9,281 Crore	₹65,700 Crore	14.1% ⁴

The "solar-ready" household segment is rapidly expanding in Tier 2 and Tier 3 cities, where discretionary spending on consumer durables surged by 72% in FY25.¹² These regions are now outpacing metro markets in adoption by 2–5X, driven by rising awareness and higher economic confidence.¹³ However, the transition from registration to installation is hampered by several structural "friction points" that require deep root cause analysis.

Root Cause Analysis: Deconstructing the Conversion

Barriers

The failure to convert 77% of registrations into installations is the result of a compounding series of hurdles. These can be categorized into financial, technical, and operational bottlenecks that alienate the consumer at various stages of the journey.

Financial Friction and the Credit-Subsidy Paradox

While the PMSGY offers generous subsidies ranging from ₹30,000 to ₹78,000 per household, the upfront cost of a 3 kW system remains significant, ranging between ₹1.8 lakh and ₹2.4 lakh.³ Even with the subsidy, a middle-income household must manage a substantial initial outlay. The current financing landscape is bifurcated between affordable but inaccessible loans and accessible but expensive credit.

Public sector banks (PSBs) offer concessional interest rates (7–8%), but their onboarding processes are often cumbersome, requiring extensive documentation and physical verification that many urban professionals find prohibitive. Conversely, NBFCs and fintech platforms like SolarSquare and Ecofy offer "instant" 10-minute digital approvals but at higher interest rates of 10–14%. This interest rate delta can significantly extend the payback period, which is typically 3–4 years under ideal conditions.¹⁴

Furthermore, the mechanism of "back-end" subsidies creates a liquidity crunch for many consumers. In the UNEP solar loan program model, subsidies were used to buy down interest rates, but in the PMSGY model, the consumer often has to finance the full amount or a large down payment (typically 30%) and wait for the subsidy to be released after commissioning.¹⁴ Any delay in the release of these funds—due to data entry errors or inspection lags—erodes consumer confidence and deters neighbors from applying.¹⁷

Supply Chain Imbalances: The DCR and ALMM Constraints

The "Domestic Content Requirement" (DCR) is a double-edged sword. While intended to foster an *Atmanirbhar* (self-reliant) solar ecosystem, it currently serves as a massive supply-side bottleneck for residential projects. To qualify for subsidies, households must use modules where both the cells and modules are manufactured in India.¹⁸

The disparity in India's manufacturing capacity is the root of the "Supply Squeeze." As of mid-2025, module manufacturing capacity reached 118 GWp, but solar cell capacity lagged at only 27 GWp, with effective operational capacity for cells as low as 11–13 GWp.¹⁸ This imbalance forces DCR-compliant module prices to remain high—between ₹23 and ₹28 per watt—nearly double the cost of non-DCR modules made with imported cells.¹

Component Category	Installed Capacity (GWp)	Effective Output (GWp)	Import Dependency (%)
--------------------	--------------------------	------------------------	-----------------------

Solar Modules	118	80–85	10–15% ¹⁸
Solar Cells	27	11–13	40–45% ¹⁸
Wafers/Ingots	<5	Negligible	>95% ¹⁸
Solar Glass	Expanding	Domestic Sourcing	High ²²

Consumers often face a choice: wait 3–4 months for expensive, subsidy-eligible DCR modules, or install cheaper, more efficient non-DCR modules (using TOPCon or HJT technology) immediately for private use.¹ Many high-consumption households in Tier 1 cities are choosing the latter, effectively opting out of the national mission to achieve faster ROI through private investment.

Operational and Regulatory Bottlenecks: The DISCOM Inertia

The most significant "Last Mile" hurdle is the local Power Distribution Company (DISCOM). The net-metering approval process is the primary point of failure in the conversion chain. Despite the national portal's attempts to streamline approvals, the ground reality is a patchwork of state-level regulations and DISCOM resistance.²³

DISCOMs have historically viewed residential solar as a revenue threat. High-paying residential consumers who switch to solar reduce the DISCOM's ability to cross-subsidize agricultural and low-income users. This has led to "passive-aggressive" delays, where net-metering approvals and physical inspections can take 45 to 120 days, far exceeding the mandated 30-day window.¹⁷ In states like Karnataka, the "Gruha Jyothi" scheme (offering 200 units of free power) has further cannibalized the solar incentive for small households, leading to a conversion rate as low as 1.1% in some districts.²⁵

Technical hurdles also persist. The transition to bi-directional meters requires skilled personnel and available inventory, both of which are in short supply in many states. Incorrect application data entries and billing misprocessing—exemplified by cases in Puducherry where consumers received bills instead of credits—further damage the reputation of the net-metering model.¹⁷

Landscaping and Market Sizing: Mapping the Potential

The Indian rooftop solar market is a bifurcated landscape where the Commercial and Industrial (C&I) segment has historically dominated, while the residential segment is currently experiencing a structural growth spurt.

Sub-Sector Dynamics: C&I vs. Residential

As of early 2025, the C&I segment accounted for approximately 60% of India's rooftop solar market

share.¹⁷ This dominance is driven by high commercial electricity tariffs and the ability of industrial entities to leverage CAPEX investments for tax benefits through accelerated depreciation.¹⁷ In contrast, the residential sector, which lagged at around 27% share for years, is now projected to grow at a CAGR of over 23% through 2031, fueled by the PMSGY subsidies.¹⁷

Market Segment	Share (2024-25)	Growth Driver	Barrier
Commercial & Industrial	~60%	High Tariffs / PPA Models	Regulatory Caps
Residential	~27%	PMSGY Subsidies / Awareness	Upfront Cost / DISCOMs
Institutional/Govt	~13%	Mandates / Sustainability	Budgetary Lags
Data Source	¹⁷	¹⁷	¹⁷

Proxy Analysis for Total Addressable Market (TAM)

Estimating the true potential for residential solar requires moving beyond census data to discretionary spending and appliance penetration metrics. Air conditioner (AC) penetration is the most robust proxy for the "solar-ready" household. A 1.5-ton AC unit can consume 1.5 to 2.5 units of electricity per hour; a household running multiple ACs during the Indian summer will inevitably cross the 300-unit monthly threshold, making solar a necessity rather than a luxury.⁸

The India AC market is expected to grow from USD 5.6 billion in 2024 to USD 16.9 billion by 2032.²⁹ This growth is most rapid in Tier 2 and Tier 3 cities, which are now witnessing an "early and severe" summer pattern.⁸ Using AC penetration data, analysts estimate that approximately 25-30 million households currently fall into the "high-intent" segment for rooftop solar, representing a TAM that far exceeds the government's 1-crore target.

Regional Leaders and Laggards

The adoption of solar is highly concentrated in a few states that have created a conducive ecosystem. Gujarat remains the undisputed leader, accounting for 30.1% of national PMSGY installations as of mid-2025.⁹ This is due to a mature vendor network, proactive DISCOMs, and the "Surya Gujarat" policy foundation.⁹ Kerala also demonstrates high conversion ratios (exceeding 65% of applications), supported by a high-literacy consumer base and efficient process management.⁴

Conversely, large states like Uttar Pradesh and Rajasthan have high registration volumes but struggle with

"vendor density." While Uttar Pradesh led the country in capacity additions in Q3 2025, its historical conversion rate has been lower due to the vast geographical spread and the difficulty of maintaining quality control across a fragmented installer base.²³

Proposed Solutions: Actionable Strategies to Bridge the Gap

Solving the conversion crisis requires a shift from a "subsidy-driven" mindset to a "customer-centric" energy solution model.

The "Finance and Subsidy Concierge" Model

The primary friction in the solar journey is cognitive load. A household must deal with the national portal, the DISCOM, the bank, and the vendor. Startups like SolarSquare have pioneered the "Full-Stack" model to eliminate these middlemen.¹⁰ The "Concierge" approach involves:

1. **Instant Digital Financing:** Providing 10-minute loan approvals using alternate credit data and IoT-based asset performance monitoring to reduce lender risk.¹⁴
2. **Embedded Subsidy Management:** The vendor takes on the responsibility of tracking the subsidy on the national portal, providing the customer with real-time updates and ensuring the application data matches DISCOM records.¹⁰
3. **GoodZero Guarantee:** Earning trust by "underwriting" the performance of the system. If the panels don't produce the promised units, the vendor fixes the issue at no cost, addressing the "fear of poor quality" prevalent in Tier 2 cities.¹⁰

Virtual Net-Metering (VNM) for Urban Dwellers

Urban India is increasingly moving toward multi-story apartments where individual roof rights are absent. Virtual Net-Metering (VNM) allows a single large solar plant (either on a common roof or off-site) to be shared among multiple residents. Raipur, Chhattisgarh, became India's first city to successfully implement residential VNM under the PMSGY.³⁴

In this model, a housing society installs a 100 kW plant. The power is fed into the grid, and the resulting "solar credits" are distributed to each flat's individual electricity bill based on their initial investment.³⁴ This democratizes solar for the "roof-less" middle class. If scaled, VNM could unlock a massive market segment currently excluded by the traditional RTS model.

Solar-EV Bundling: The ROI Multiplier

The rapid rise of Electric Vehicles (EVs) in India (reaching 7.6% of sales in 2025) presents a unique opportunity to bundle "Zero-Fuel" solutions.³⁵ The economic synergy is undeniable:

- **Fuel Savings:** Charging a 30 kWh EV battery at home costs ~₹180 (domestic tariff) vs. ~₹1,800 for an equivalent petrol trip.³⁶
- **Solar ROI:** By using solar energy to charge an EV, the household effectively increases its self-consumption of solar power, which is more lucrative than selling power back to the grid at the

lower wholesale rate (Net Billing).³⁶

- **Real Estate Value:** Properties marketed as "EV-Solar Ready" are already commanding a 3–5% price premium in cities like Bengaluru and Delhi.³⁶

Policy Recommendations for Streamlined Conversion

To accelerate the 22.7% conversion rate, a series of regulatory "grease points" are required.

- **Standardized Interconnection:** The Ministry of Power should mandate a "Deemed Approval" status for residential projects if the DISCOM fails to inspect the site within 15 days of installation.
- **DCR Reservation:** To prevent the DCR cell shortage from stalling PMSGY, the government could reserve 20% of domestic cell output specifically for residential projects, ensuring a predictable supply for small-scale installers.¹
- **Direct Benefit Transfer (DBT) of Interest:** Rather than back-end capital subsidies, the government could explore interest-free solar loans for the first 5 years, with the subsidy amount directly paying down the EMI.¹⁶

Landscaping the Competitive and Funding Ecosystem

The "ClimateTech" sector in India is undergoing a structural shift toward "Execution Capital." In H1 2025, climate startups raised USD 1.28 billion, more than double the amount in the same period in 2024.³⁸

Funding Trends and Investor Sentiment

The investment landscape is "barbell-shaped," with significant capital flowing to mature, integrated players while early-stage startups face a "Series A gap".³⁸ Investors are prioritizing "Full-Stack" companies that control the customer experience.

- **SolarSquare:** Raised USD 40 million in Series B funding led by Lightspeed, reflecting confidence in its full-stack residential model.¹⁰
- **Ecofy:** Focusing on "Green Loans" for solar and EVs, bridging the financing gap for retail consumers who don't fit traditional bank profiles.⁴⁰
- **Manufacturing Integration:** Investors are rewarding companies that integrate across the value chain. Waaree and Premier Energies have seen successful IPOs, with valuations driven by their manufacturing depth rather than just project volume.⁴¹

The Emergence of the SME IPO as an Exit Path

A notable shift in 2024-25 is the use of the SME Exchange as a credible exit and growth path for regional solar players. Companies like Alpex Solar and Sahaj Solar have demonstrated that liquidity is possible for EPC firms once they reach approximately ₹300 crore in sales.⁴¹ This provides a "Series C" alternative for venture funds, allowing for faster exit cycles and more recycled capital within the ecosystem.

Listed Company	Market Cap (Valuation)	Revenue Multiple (P/S)	Integration Level
Waaree Energies	High	6x	Fully Integrated ⁴¹
Premier Energies	High	6.8x	Cell + Module ⁴¹
Sahaj Solar	SME	2-3x	Regional EPC ⁴¹
Ola Electric	High	5x	EV + Battery ⁴¹

Future Outlook: The Next 5–10 Years (2025–2034)

The trajectory of the Indian residential solar market points toward a shift from "net-metered solar" to "smart energy hubs."

Battery Energy Storage System (BESS) Integration

The intermittent nature of solar and the rising peak demand in the evenings make battery storage inevitable. As BESS costs fall, the market will transition to hybrid systems. The 2026-27 Union Budget provided a massive boost by increasing Viability Gap Funding (VGF) for BESS ninefold, from ₹100 crore to ₹1,000 crore.⁴² This will enable households to achieve 24/7 grid independence and protect themselves against "Time-of-Day" (ToD) tariffs, which are being rolled out nationwide by 2025.²⁸

Peer-to-Peer (P2P) Energy Trading and Virtual Power Plants

The digitalization of the grid will eventually allow households to trade excess solar power with their neighbors. Trials for blockchain-based P2P trading have already been initiated in Uttar Pradesh and Karnataka.²⁸ This "Uber-ization" of electricity turns the household from a passive consumer into a "prosumer," capable of generating revenue during the day and drawing power at night.

The Decoupling of Subsidies and the Rise of "Solar-as-a-Service"

As system costs reach "grid parity"—the point where solar electricity is cheaper than grid power without subsidies—the market will shift toward the RESCO (Renewable Energy Service Company) model for residential clusters. In this model, the consumer pays zero upfront cost and merely pays a per-unit fee to a third-party developer who owns and maintains the panels.¹⁷

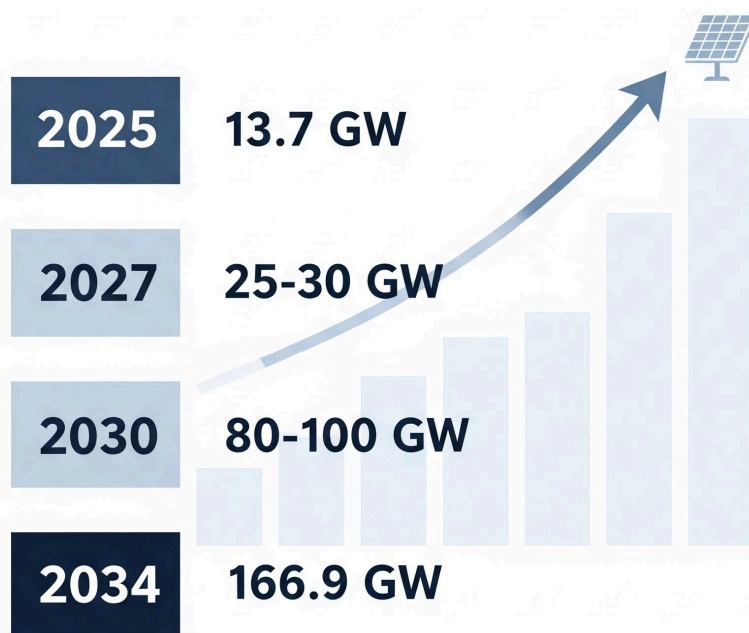
Projected Market Trajectory

By 2034, the India solar rooftop market is expected to reach a volume of 166.87 GW, growing at a CAGR of 28.4% from 2025 levels.²⁸ The residential segment will be the primary driver of this volume, as AC

penetration and EV ownership become universal in urban and peri-urban India.

Year	Projected Installed Volume (GW)	Technology Milestone	Policy Tailwind
2025	13.7 GW	TOPCon Adoption	PMSGY Full Rollout ²⁸
2027	25-30 GW	BESS Parity	1 Crore Home Deadline ¹
2030	80-100 GW	V2H Implementation	500 GW RE Target ⁴⁵
2034	166.9 GW	Smart Grid/P2P Trading	Carbon Market Maturity ²⁸

Projected Installed Volume (GW)



Conclusions and Strategic Recommendations

The conversion gap in India's residential solar market is a systemic bottleneck that requires a coordinated "pincer movement" from policy, technology, and finance. The following synthesis offers a roadmap for stakeholders:

1. **For Policy Makers:** The focus must shift from "registration targets" to "commissioning timelines." Mandating automated net-metering approvals and penalizing DISCOM delays will be the single most effective intervention to unlock the 5.8 million pending applications.
2. **For Financial Institutions:** There is a critical need for "low-documentation, high-speed" credit. By integrating IoT data from solar inverters into credit scoring models, banks can offer instant loans that mirror the ease of consumer durable financing.
3. **For Solar Startups:** The "Concierge" model is no longer optional. To survive in the "trust deficit" Tier 2/3 markets, vendors must own the permitting and performance risks, offering "Full-Stack" solutions that include maintenance and EV integration.
4. **For DISCOMs:** Rather than resisting solar, DISCOMs must pivot to becoming "Grid Facilitators." By leveraging rooftop solar to manage peak demand and reduce AT&C losses (which rose to 16.12% in FY24), DISCOMs can turn a revenue threat into an operational asset.⁴⁶

The mathematical reality of India's solar potential is clear. The annual generation from the targeted 30 GW capacity can be derived as:

$$E = P \times 8760 \times CUF$$

Where P is 30,000,000 kW and CUF (Capacity Utilization Factor) is approximately 19%. This yields over 49.9 billion units of clean electricity annually.¹¹ Capturing this "free" energy from the sun is not just an environmental imperative; it is the most potent financial instrument available for the empowerment of the Indian household. As the conversion gap closes, India will move from a nation of "solar registrants" to a global leader in "distributed energy independence."

Works cited

1. Residential rooftop solar grows under PM Surya Ghar Yojana but ..., accessed February 18, 2026, <https://ieefa.org/resources/residential-rooftop-solar-grows-under-pm-surya-ghar-yojana-gaps-persist>
2. Putting just transition at the heart of solar power, accessed February 18, 2026, https://re-course.org/wp-content/uploads/2024/10/Banking-on-Renewables_case-study-India.pdf
3. Over 28 lakh homes powered by rooftop solar under PM Surya Ghar Scheme: Govt in Parliament - Organiser, accessed February 18, 2026, <https://organiser.org/2026/02/05/338535/bharat/over-28-lakh-homes-powered-by-rooftop-solar-under-pm-surya-ghar-scheme-govt-in-parliament/>
4. PM Surya Ghar Hits 4.9 GW, But Conversion and Finance Challenges Persist: IEEFA, accessed February 18, 2026, <https://www.saurenergy.com/solar-energy-news/pm-surya-ghar-hits-49-gw-but-conversion-and-finance-challenges-persist-ieee-10559738>

5. India power sector review 2025: Record clean energy deployment drives historic decline in coal generation, accessed February 18, 2026,
https://energyandcleanair.org/wp/wp-content/uploads/2026/01/CREA_India-Power-Sector-Review-2025.pdf
6. India's Latest Budget Reveals New Details for National Energy Transition, accessed February 18, 2026,
<https://www.energypolicy.columbia.edu/indias-latest-budget-reveals-new-details-for-national-energy-transition/>
7. India Climate Tech Startup Landscape, accessed February 18, 2026,
https://www.bk.mufg.jp/global/globalnetwork/asiaoceania/pdf/India_Climate_Tech_Startup_Landscape.pdf
8. India Air Conditioner Market Size and Share Analysis - Growth Trends and Forecast Report 2025-2033, accessed February 18, 2026,
<https://www.researchandmarkets.com/reports/5758501/india-air-conditioner-market-size-share>
9. Pradhan Mantri Surya Ghar Yojana: India's Rooftop Solar Scheme ..., accessed February 18, 2026,
<https://jmkresearch.com/renewable-sector-published-reports/advancing-residential-rooftop-solar-adoption-in-india-under-pm-surya-ghar-yojana/>
10. How SolarSquare has a chance to become a home energy brand in ..., accessed February 18, 2026,
<https://www.forbesindia.com/article/news/how-solarsquare-has-a-chance-to-become-a-home-energy-brand-in-india/94835/1>
11. PM Surya Ghar: More Than a Climate Scheme - BlueKraft Digital Foundation, accessed February 18, 2026,
https://www.bluekraft.in/wp-content/uploads/2025/06/Jaya_Dubey.pdf
12. India Household Appliances Market | Industry Report, 2030 - Grand View Research, accessed February 18, 2026, <https://www.grandviewresearch.com/industry-analysis/india-household-appliances-market-report>
13. Tier-2 Cities Drive India's Rooftop Solar Boom, Outpacing Metros by ..., accessed February 18, 2026,
<https://www.energetica-india.net/news/tier-2-cities-drive-india-rooftop-solar-boom-outpacing-metros-by-up-to-5x-solarsquare>
14. Everything You Should Know About Solar Financing - SolarSquare, accessed February 18, 2026,
<https://www.solarsquare.in/blog/solar-financing/>
15. Financing by SolarSquare: 5-year (60-month) EMI scheme Launched - Rooftop Solar for your home and more - Blog, accessed February 18, 2026,
<https://www.solarsquare.in/blog/financing-by-solarsquare-5-year-60-month-emi-scheme-launched/>
16. CASE STUDY 12: INDIA – SOLAR LOAN PROGRAMME - energypedia, accessed February 18, 2026,
https://energypedia.info/images/5/53/India_UNEP_Solar_Loan_Programme.pdf
17. India Rooftop Solar Market Report, Government, Residential, Commercial, Major ... - Nexdigm, accessed February 18, 2026,
<https://www.nexdigm.com/market-research/report-store/india-rooftop-solar-market-report/>
18. Solar Equipment: Localisation at the Forefront - CareEdge | Ratings, accessed February 18, 2026,
<https://www.careratings.com/report/solar-equipment-localisation-at-the-forefront>
19. DCR vs Non-DCR Solar Panels (2025 Guide): Cost, Efficiency & Policy Explained, accessed February 18, 2026,
<https://heavengreenenergy.com/dcr-vs-non-dcr-solar-panels-2025-guide-cost-efficiency-policy-explained/>
20. Mandating DCR Modules Could Raise Solar Tariffs by 30–40 Paise/Unit: CareEdge, accessed February 18, 2026,
<https://www.saurenergy.com/solar-energy-news/mandating-dcr-modules-could-raise-solar-tariffs-by-3040-paise-unit-careedge-10503815>
21. India Builds 44 GW Solar Module Manufacturing Capacity In H1 2025 - TaiyangNews, accessed February 18, 2026,
<https://taiyangnews.info/markets/india-builds-44-gw-solar-module-manufacturing-capacity-in-h1-2025>
22. India's Union Budget 2026 Boosts Solar Allocation By 32% - TaiyangNews, accessed February 18, 2026,
<https://taiyangnews.info/markets/india-union-budget-2026-boosts-solar-allocation-by-32-percent>
23. Talk is Cheap: Delhi's Discoms Fall Short on Solar Ambitions - Saur Energy, accessed February 18, 2026,
<https://www.saurenergy.com/solar/talk-is-cheap-delhis-discoms-fall-short-on-solar-ambitions-11099127>
24. Bridging the Rooftop Divide: Reforming Solar Net Metering in India, accessed February 18, 2026,
<https://www.cpnr.in/articles/reforming-solar-net-metering-in-india>
25. Rooftop solar rollout under PM Surya Ghar scheme slows down in Karnataka - CarbonCopy, accessed February 18, 2026,

- <https://carboncopy.info/rooftop-solar-rollout-under-pm-surya-ghar-scheme-slows-down-in-karnataka>
26. Commercial Solar Terms Glossary | Big Sun Solar, accessed February 18, 2026, <https://www.bigsunsolar.com/solar-terms-glossary>
 27. India Solar Energy Market Size, Share & Growth 2031 - Mordor Intelligence, accessed February 18, 2026, <https://www.mordorintelligence.com/industry-reports/india-solar-energy-market>
 28. India Solar Rooftop Market Outlook Report 2025-2034: - GlobeNewswire, accessed February 18, 2026, <https://www.globenewswire.com/news-release/2025/11/11/3185388/28124/en/India-Solar-Rooftop-Market-Outlook-Report-2025-2034-Expansion-Fueled-by-Avaada-Waaree-Energies-Tata-Power-Renewable-Energy-and-Adani-through-Digitalization-and-Integrated-Energy-So.html>
 29. India Air Conditioner Market Size, Share and Growth Report 2032 - Credence Research, accessed February 18, 2026, <https://www.credenceresearch.com/report/india-air-conditioner-market>
 30. mercom india rooftop solar market report q3 2025 – 4.9 gw installed in 9m 2025, accessed February 18, 2026, <https://www.mercomindia.com/product/rooftop-solar-market-report-q3-2025>
 31. Why SolarSquare Pivoted To Residential #shorts #solar #electricity - YouTube, accessed February 18, 2026, <https://www.youtube.com/shorts/y6z9Rv9BJCk>
 32. Financing India's Climate Tech Ecosystem - Saur Energy, accessed February 18, 2026, <https://www.saurenergy.com/solar-energy-blog/bridging-the-funding-gap-in-indias-climate-tech-ecosystem-9773985>
 33. Why Tier-2 and Tier-3 cities will drive India's next big solar jump - pv magazine India, accessed February 18, 2026, <https://www.pv-magazine-india.com/2025/12/23/why-tier-2-and-tier-3-cities-will-drive-indias-next-big-solar-jump/>
 34. Chhattisgarh: India's first virtual net metering under PM Surya Ghar Yojna takes off successfully - ANI News, accessed February 18, 2026, <https://www.aninews.in/news/national/general-news/chhattisgarh-indias-first-virtual-net-metering-under-pm-surya-ghar-yojna-takes-off-successfully20251015221804/>
 35. How Real Estate Developers Can Monetize EV Charging in India (2026 Guide) - Bolt.Earth, accessed February 18, 2026, <https://bolt.earth/blog/how-real-estate-developers-can-monetize-ev-charging-in-india-2026-guide>
 36. The Rise of EV-Ready Homes in India: Why Future-Proofing Your Property Matters, accessed February 18, 2026, <https://bolt.earth/blog/the-rise-of-ev-ready-homes-in-india-why-future-proofing-your-property-matters>
 37. Net Billing vs. Net Metering For Solar Overproduction - Paradise Energy, accessed February 18, 2026, <https://www.paradisepowerenergy.com/blog/net-billing-vs-net-metering-for-solar/>
 38. The State of Climate Tech Investment in India: Big Rounds In, Small Ones Vanishing, accessed February 18, 2026, <https://www.finetrain.com/blog/the-state-of-climate-tech-investment-in-india-big-rounds-in-small-ones-vanishing/>
 39. SolarSquare Secures \$40 Million Series B to Expand India's Rooftop Solar Revolution, accessed February 18, 2026, <https://climateinsider.com/2024/12/12/solarsquare-secures-40-million-series-b-to-expand-indias-rooftop-solar-revolution/>
 40. Climate-Tech Funding in India: Beyond Seed Funding—Early-Stage Gaps & How to Bridge them. | Ecofy Finance, accessed February 18, 2026, <https://www.ecofy.co.in/blogs/climate-tech-funding-india-beyond-seed-funding-early-stage-gaps-how-bridge-them>
 41. Public Valuations Are Setting The New Climate Startup Valuation Benchmarks - FineTrain, accessed February 18, 2026, <https://www.finetrain.com/blog/public-valuations-are-setting-the-new-climate-startup-valuation-benchmarks/>
 42. Union Budget Increases VGF Support for BESS, Hikes PM Surya Ghar Allocations, accessed February 18, 2026, <https://www.saurenergy.com/solar-energy-news/union-budget-increases-vgf-support-for-bess-hikes-pm-surya-ghar-allocations-11064063>
 43. Budget Allocates INR 22,000 Cr for PM Surya Ghar, INR 1,000 Cr for BESS VGF, INR 500 Cr for CCUS - Asia Pacific | Energetica India Magazine, accessed February 18, 2026,

<https://www.energetica-india.net/news/budget-allocates-inr-22000-cr-for-pm-surya-ghar-inr-1000-cr-for-bess-vgf-inr-500-cr-for-ccus>

44. Community Solar: Harnessing the Sun without the Financial Burden, accessed February 18, 2026, <https://ornatesolar.com/blog/community-solar-harnessing-the-sun-without-the-financial-burden>
45. India's Solar Supply Chain in 2025: Opportunities and Challenges, accessed February 18, 2026, <https://rayzonsolar.com/blog/india-solar-supply-chain-2025>
46. Reform Track: Discoms make progress, but challenges remain - Power Line Magazine, accessed February 18, 2026, <https://powerline.net.in/2025/10/30/reform-track-discoms-make-progress-but-challenges-remain/>
47. Net Metering – SEIA - Solar Energy Industries Association, accessed February 18, 2026, <https://seia.org/net-metering/>